CASE-STUDY PRACTICE | CAPSTONE 8 | STUDY EDITION

Feed Ranking: From a Prize Metric to an Ordering

Scenario questions on ranking the titles in a personalized row, each opening into deeper follow-ups, with answers that explain how to think. Built on the AI System Design Primer case study of the same name.

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HOW TO STUDY WITH THIS BOOK

1. Read the amber **concepts box** at the start of a part and look up anything unfamiliar. 2. Read a **scenario** and, before reading further, write down what you would ask and what you would check. 3. Read the green **how to think** box, which names the mental move the question tests. 4. Walk down the nested steps: cover each answer, predict it, then compare. Each answer exposes the next question. The last step of every question asks for the general lesson in one sentence.

The pattern behind most questions

- A number or a technique arrives with authority: a published benchmark, a cost estimate, a shard count, a rate.
- The interviewer asks what it assumes, and what happens when the assumption fails.
- A strong answer names the hidden assumption, puts a number on it, and says how to measure it on your own data.
- The last step usually reframes the original question.

EVIDENCE TAGS

source-confirmed: checked against the named primary source. **illustrative**: this document's planning assumption. **recomputed**: arithmetic re-derived from the stated inputs. **reasoned**: an argument from the design, not tested.

A NOTE ON SOURCES AND ON WHAT WAS CHECKED

The scenarios are written for this document; none comes from a real company question bank. Nothing here was run on data, a model or a serving system: figures are the posts' or arithmetic from stated inputs, and homepage sizes, weights, test sizes and request rates are illustrative. The last page lists the sources and says how much of each was read. A separate Mock Interview edition (answers hidden) uses the same questions and numbering.

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The case in one page

A summary of the case study to read before the questions.

Problem	Order the titles in each row of a member's homepage so that the ones the member is most likely to play and enjoy come first, in real time.
Core idea	Rating prediction was a proxy. The ranking function combines predicted rating with popularity and other features, and is learned from historical positive and negative examples.
Objective	Member satisfaction and month-to-month retention, which correlates with consumption of video (source-confirmed).
Scale in 2012	More than 23 million subscribers, 2 billion hours in a quarter, 4 million ratings and 2 million queue adds a day, 75% of viewing from recommendations (source-confirmed).
What shipped	SVD and RBM, not the Grand Prize ensemble, whose gains did not justify the engineering effort (source-confirmed).
Evidence	Prize error figures, one function form, and a bar chart with no stated metric: about a third for adding ratings, about 225% for more features and tuning.
Not published	Latency, request rate, catalog size, weights, the features behind the large gain, and any serving architecture.
Easy-to-miss traps	A better RMSE can give a worse ordering. Hours per play from the posts' figures is 2.4 or more. The chart names no metric. Twenty cells give a 62% chance of a false winner.

Numbers to know

Each figure carries the tag that says how it was obtained. The posts give a few exact volumes and a function form but no serving figures, so homepage sizes, weights and test sizes are illustrative.

Quantity	Value	Evidence
Prize figures	Cinematch 0.9525; target 0.8572 (10%); SVD 0.8914 (6.4%); RBM 0.8990 (5.6%); blend 0.88 (7.6%); first Progress Prize 8.43%	source-confirmed; percentages recomputed
2012 scale	More than 23 million subscribers, 47 countries; 2 billion hours in Q4 2011; 2 million queue adds and 4 million ratings a day; more than 5 billion ratings; 75% of viewing from recommendations	source-confirmed (Part 1)
Derived rates	4,000,000 ÷ 86,400 = 46 ratings a second; 21.7 million hours a day; 57 minutes per subscriber a day; 217 ratings on file per subscriber	recomputed
Hours per play	21.7 million hours ÷ 3 million plays = 7.2 h; ÷ 9 million = 2.4 h	recomputed
Homepage load	40 × 75 = 3,000 scores; 34.5 million loads a day; about 400 a second, 1,600 at peak; 4.8 million scores a second	illustrative; recomputed
Popularity spike	(167 × 100 + 10,000) ÷ 168 = 159 an hour; the spike is 63 times the weekly average	illustrative; recomputed
RMSE against order	Model 1: RMSE 0.32, order B A C; Model 2: RMSE 0.71, order A B C	illustrative; recomputed
Weight flip	A (0.3, 4.5) against B (4.6, 3.6): A wins only if w2 ÷ w1 is above 4.8	illustrative; recomputed
Chart	Ratings about one third (about 32% by reading the bar); more features and optimized models about 225%; metric not stated	source-confirmed (figure); read from the bar
MRR and FCP	Ranks 1, 3, 10: MRR 0.48; order B A C gives FCP 2 ÷ 3	illustrative; recomputed
Test precision	5,000 members a cell at 90% retention: standard error 0.42%; margin on a difference 1.2 points	illustrative; recomputed
Many cells	1 − 0.95^19 = 62% for 20 cells; 19% for 5 cells	recomputed
Tests	Thousands of members, 2 to 20 cells, scores of parallel tests; Top10 Marathon: 10 weeks, 6 ideas a week	source-confirmed (Part 2)

Decision table

What this design chooses, and when to choose otherwise.

Decision	Default here	Choose otherwise when	Evidence
Objective	Satisfaction and retention, with consumption as the correlate	Rating accuracy only if recommendations are not ranked	source-confirmed
Ranking function	A learned combination of popularity, predicted rating and more features	Popularity alone if personalization is not wanted	source-confirmed
Setting weights	Learn them from historical positive and negative examples	Sampling weights through A/B tests, which the post calls costly	source-confirmed
Popularity	Several windows and groups	One global number if trends and regions do not matter	source-confirmed
Offline metrics	A basket: ranking, classification, RMSE and diversity	A single metric for a quick check	source-confirmed
Final judgment	A/B test on engagement and retention	Offline metrics only to choose what to test	source-confirmed
Many cells	A threshold and a confirmation step fixed in advance	Fewer cells	reasoned
Serving	Cache shared signals, compute personal ones per request	Precompute whole rows only if freshness is not needed	reasoned

PART 1

Framing the problem: proxies, objectives and the sizing

Why a prize metric was only a proxy, why a better error score can give a worse ordering, and which figures can be combined.

READ THESE FIRST: CONCEPTS COVERED IN THIS PART

RMSE	Root mean squared error: the square root of the mean of the squared differences between predicted and actual values. Large misses count more, and each prediction is scored alone.
Proxy metric	A measurable stand-in for something you cannot measure directly. It can improve while the real goal does not.
Ranking	Choosing the order of items in a list. Here, which titles come first in a row, for one member, in real time.
Predicted rating r(u, v)	The stars a model expects member u to give title v. One input to the ranking function.
Retention	Whether a member keeps subscribing from month to month. The posts' business objective, along with satisfaction.
Stream play	One logged start of viewing. The posts do not define it, and it may stand for a session of several titles.
Cross-check	Combining two stated figures to see whether they are compatible, for example hours streamed against plays.
Scoring, sorting and filtering	The three operations the posts say ranking is decomposed into: compute a score for each title, order by it, and drop ineligible titles.
Where to read	Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 1 (April 2012) | Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 2 (June 2012)

01 What did the prize buy, and what shipped?

FRAMING

SCENARIO

In 2006 Netflix offered 1 million dollars for a 10% improvement in rating prediction. Years later, a colleague says the winning ensemble must be running in production, and that the prize shows rating accuracy was the goal. Check both claims against the posts.

HOW TO THINK ABOUT IT

Work out the improvement of each model over the baseline, then read what the posts say shipped and why.

Q What was the proxy?

A Netflix needed a question that was easier to evaluate than better recommendations, so it chose the root mean squared error of predicted ratings. The baseline, Cinematch, scored 0.9525, and the target was 0.8572 or less (source-confirmed).

- 10% of 0.9525 is 0.0953, which gives 0.8572.

Q How far did the pieces get?

A SVD alone scored 0.8914, a 6.4% improvement. RBM alone scored 0.8990, 5.6%. A linear blend of the two scored 0.88, 7.6%. The first Progress Prize went to a team with an 8.43% improvement and a combination of 107 algorithms after more than 2,000 hours of work.

- The percentages are arithmetic on the posts' figures.

Q What shipped?

A The two underlying algorithms, SVD and RBM, after overcoming limits: they were built for 100 million ratings and Netflix had more than 5 billion, and they were not built to adapt as members added ratings. The Grand Prize ensemble of hundreds of models was evaluated offline and not shipped, because the added accuracy did not justify the engineering effort.

- The post says the two algorithms are still used as part of the recommendation engine.

Q Was accuracy the goal?

A No. The post says the prize objective is just one of many components, and lists context, popularity, interest, evidence, novelty, diversity and freshness. Part 2 restates the goal as the probability that a member chooses to watch a title and enjoys it enough to come back.

- Streaming also changed what data exists: Netflix can observe whether a video was watched fully or partly.

Q Reframe it.

A A competition metric is a proxy chosen to be easy to score. Say what shipped and what did not, and what the real objective was.

Basis: The figures and statements are from Part 1 and Part 2 (source-confirmed); the percentages are arithmetic (recomputed).

02 A better error score and a worse ordering

METRIC

SCENARIO

Two rating models are compared on three titles that a member would truly rate A 4.5, B 4.0 and C 3.0. Model 1 predicts 4.0, 4.2 and 3.1. Model 2 predicts 5.0, 3.5 and 2.0. A manager picks the model with the better RMSE. Show what the choice does to the row.

HOW TO THINK ABOUT IT

Compute RMSE for each, then the order each produces.

Q What are the errors?

A Model 1: 0.5, 0.2 and 0.1. Model 2: 0.5, 0.5 and 1.0.

- Each is a prediction's distance from the truth.

Q What are the RMSE values?

A Model 1: sqrt((0.25 + 0.04 + 0.01) ÷ 3) = sqrt(0.10) = 0.32. Model 2: sqrt((0.25 + 0.25 + 1.00) ÷ 3) = sqrt(0.50) = 0.71.

- Model 1 has less than half the error.

Q What are the orders?

A Model 1 ranks B first, then A, then C. The top pair is swapped. Model 2 ranks A, B, C, which is correct.

- Model 2 has the larger error and the right order.

Q Why does that happen?

A RMSE scores each prediction alone and has no notion of order across titles. Model 1 is close everywhere and misses the gap between A and B. Model 2 is far off in places and keeps the order.

- A pair-based measure such as the fraction of concordant pairs gives 2 of 3 pairs for Model 1 and 3 of 3 for Model 2.

Q Reframe it.
A Choose the metric that matches the use. For a list, measure the order, and keep the pointwise error as one of several signals.

Basis: The distinction between RMSE and ordering follows from the posts' discussion of ranking against rating prediction (source-confirmed); the three titles and both models are invented; arithmetic recomputed.

03 Can you combine the posts' scale figures?

ESTIMATION

SCENARIO

Part 1 says subscribers streamed 2 billion hours in the last quarter of 2011, and Netflix has more than 23 million subscribers. Part 2 says several million stream plays a day. An analyst divides one by another. Check which divisions make sense.

HOW TO THINK ABOUT IT

Convert to daily figures, then divide, and see what each result implies.

Q What is the daily viewing?
A 2,000,000,000 hours ÷ 92 days = about 21.7 million hours a day. Per subscriber: 21.7 million ÷ 23 million = 0.95 hours, about 57 minutes (source-confirmed inputs).

- The subscriber count is from April 2012 and the hours are for the previous quarter, so it is approximate.

Q What does hours per play give?
A With 3 million plays a day, 7.2 hours per play. With 9 million, 2.4 hours per play.

- Even at nine million plays the average is longer than most films.

Q What does that mean?
A Either a play stands for a session of several titles, which the posts say members do, or several million is loose. The two figures cannot be combined until a play is defined.

- The posts do not define it.

Q Which figures can be used?
A The daily ratings: 4,000,000 ÷ 86,400 = about 46 a second. And 5 billion ratings ÷ 23 million subscribers = about 217 each.

- These divide cleanly because both count the same thing.

Q Reframe it.
A Check that two figures count the same unit before you divide one by the other. A result that looks wrong is a clue to a definition.

Basis: The figures are from Part 1 and Part 2 (source-confirmed); arithmetic recomputed; the reading of a play is reasoned.

04 Size the scoring load

ESTIMATION

SCENARIO

A homepage has rows of titles, and each title in a row is scored for the member. The posts give no catalog size or number of rows. Make a stated assumption, size the load, and say where the real cost is.

HOW TO THINK ABOUT IT

Multiply rows by titles, then by loads a second, then count what each score reads.

Q What are the assumptions?

A 40 rows with 75 titles each, 23 million subscribers loading the homepage 1.5 times a day, and a peak of 4 times the average (all illustrative). Part 1 says genres are chosen from the thousands possible.

- None is in the posts.

Q What is the load?

A 40 × 75 = 3,000 scores per homepage. 23,000,000 × 1.5 = 34.5 million loads a day, 400 a second on average and 1,600 at peak. 1,600 × 3,000 = about 4.8 million scores a second at peak.

- Each score is two multiplications and two additions in the posts' simple function.

Q Is the arithmetic the cost?

A No. 4.8 million scores of four operations is under 20 million operations a second, a tiny fraction of one core.

- The cost is reading the inputs: a popularity value for each title and a predicted rating for each member and title.

Q What would change the picture?

A A richer model. The posts say they tried many more features, and a model that needs hundreds of lookups per title changes the arithmetic and the reads.

- Measure reads per score before you decide where to optimize.

Q Reframe it.

A Cheap arithmetic does not make a cheap system. Count the data reads each operation needs.

Basis: The statement about genres is from Part 1 (source-confirmed); the rows, titles, loads and peak ratio are illustrative; arithmetic recomputed.

PART 2

Data and labels: signals, windows, households and exposure

What the posts list as inputs, why one popularity number is not enough, what an account is not, and how a shown title differs from a chosen one.

READ THESE FIRST: CONCEPTS COVERED IN THIS PART

Implicit signal	A signal inferred from behavior, such as a play or a queue add, not stated by the member. It is plentiful and noisy.
Popularity window	The time range over which popularity is counted, such as an hour, a day or a week. A short window follows trends and a long one is steady.
Segment popularity	Popularity counted within a group of members, such as a region, instead of over everyone.
Presentation	A recommendation as it was shown: which items, where, and how the member interacted with them.
Position bias	A title shown first is chosen more because it was first, not only because it is better.
Household	An account can be used by several people with different tastes, so its history is a mixture.
Feature normalization	Putting signals on comparable scales before combining them, so that none dominates only because of its units.
Cold start	A new member or title with no history, for whom history-based signals are empty.
Where to read	Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 1 (April 2012) | Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 2 (June 2012)

05 Whose behavior is it?

LABELS

SCENARIO

The ranking signal is mostly behavioral: plays, queue adds, searches and how members interacted with what was shown. A teammate treats each account's history as one person's taste. Explain the problem and what it means for labels.

HOW TO THINK ABOUT IT

Read what the posts say about the household and about implicit signals, then follow one account.

Q What do the posts list?

A Ratings, popularity, stream plays with duration, time of day and device, queue adds, metadata, presentations, social data, search terms and external data such as box office and reviews (source-confirmed).

- Most are implicit. Only the ratings are explicit.

Q What do they say about households?

A Part 1 says personalization is intended to handle a household that is likely to have different people with different tastes. The Top 10 row is the best guess at what everyone in the household would enjoy.

- A single-person household still has a range of interests and moods.

Q What follows for labels?

A A play is evidence about the account, and not about one person. A parent's documentary and a child's cartoon are the same member. A model that treats it as one taste averages two.

- The posts say Netflix optimizes for diversity as well as accuracy in many parts of the system.

Q What does an implicit signal not say?

A That the member liked it. A play shows they started a title. Duration shows more, since the posts say Netflix can observe whether a video was watched fully or partly.

- Prefer signals that reflect completion, and treat a short play as weak evidence.

Q Reframe it.

A Ask whose behavior a log records. When one account is several people, build for the account and do not pretend it is one person.

Basis: The data list, the household statement and the observation about partial viewing are from the posts (source-confirmed); the reading of what follows for labels is reasoned.

06 One popularity number, or several?

FRESHNESS

SCENARIO

A product owner asks for one all-time popularity score per title, which is simpler. The posts compute popularity over several windows and within groups. Show what the single number misses.

HOW TO THINK ABOUT IT

Compare a weekly average with a one-hour spike for one title, then think about regions.

Q What do the posts do?

A They compute popularity over various time ranges, such as hourly, daily or weekly, or group members by region or other similarity measures and compute popularity within the group (source-confirmed).

- The post does not say which windows production uses.

Q What does a weekly figure miss?

A A title that normally gets 100 plays an hour and gets 10,000 in one hour. Averaged over a week of 168 hours: (167 × 100 + 10,000) ÷ 168 = about 159 an hour. The spike is 63 times larger than the average.

- The weekly number moves a little, and the hourly number moves a lot.

Q What does a global number miss?

A Local tastes. A title can be popular in one region and unknown in another, and a global count hides both.

- Grouping members by region or similarity gives each group its own signal.

Q What does it cost?

A More computations: one per window per group per title. And a choice of which to use, which can itself be learned.

- A short window is noisier, and a long one is slower to react.

Q Reframe it.

A Popularity depends on a time range and a population. Pick both for the decision, and do not assume one number fits.

Basis: The windows and groups are from Part 2 (source-confirmed); the 100 and 10,000 plays are invented; arithmetic recomputed.

07 Shown is not chosen

EXPOSURE

SCENARIO

The posts say Netflix logs presentations: what was recommended, where it was shown, and how the member interacted. A teammate trains on plays as if every title had the same chance to be played. Explain the bias.

HOW TO THINK ABOUT IT

Say what the log records, then what a title in the first position has that another does not.

Q What does the log hold?

A What was shown and where, and the member's scrolls, mouse-overs, clicks and time on the page. Netflix can look at how a recommendation decision affected the member's actions (source-confirmed).

- It records exposure, which a play log alone does not.

Q What is the bias?

A A title in the first position is seen by more members and clicked more because it was first. Training on plays alone teaches the model that whatever the old system put first is good.

- This is reasoning, and the posts do not discuss position.

Q How can the log help?

A By including position as a feature in training and setting it to a fixed value when ranking, or by weighting examples by their chance of being seen.

- The posts do not say Netflix does this.

Q What would you check?

A Play rate by position for the same title, and whether the model's scores follow the old system's order more than members' choices.

- A random-order bucket, as another system in this series uses, gives unbiased data at some cost.

Q Reframe it.

A A log of choices is also a log of what the old system offered. Record the offer, and model it.

Basis: The presentations data is from Part 2 (source-confirmed); the bias and the remedies are reasoned and untested, and the posts do not discuss them.

08 Review the row ranker

CODE REVIEW

SCENARIO

A colleague writes the row ranker below and says it implements the posts' scoring, sorting and filtering. Every member gets nearly the same order, and rows sometimes come back with fewer titles than requested. Find the faults.

```
def rank_row(candidates, w1, w2, top_n):
    ranked = sorted(
        candidates,
        key=lambda c: w1 * c["plays_last_week"] + w2 * c["predicted_rating"],
        reverse=True,
    )[:top_n]
    return [c["title_id"] for c in ranked if not c["already_watched"]]
```

HOW TO THINK ABOUT IT

Compare the sizes of the two signals, then follow the order of the sort, the slice and the filter.

Q Why is everyone's order alike?

A The popularity input is a play count, in the thousands or millions, and the predicted rating is at most 5. With any similar weights the count decides the order for everyone.

- The posts say popularity alone produces the same ordering for every member.

Q What is the fix for the scales?

A Put both signals on comparable scales before combining them, for example a popularity percentile mapped to 0 to 5, or a log of the count, and then learn or set the weights.

- Question 9 shows how much the ratio of the weights matters.

Q Why short rows?

A It slices to the top n, and then removes watched titles. Every removed title leaves the row shorter.

- Filter first, then take the top n.

Q What else is missing?

A A tie rule. Equal scores come back in input order, which can favor whatever the candidate source listed first.

- Break ties by a stable key, or by a second signal.

Q State the fix, and a test.

A Normalize popularity, filter, sort, then slice. Test: two members with different predicted ratings get different orders when popularity is equal, and a row with 3 watched titles of 20 still returns n titles.

- Also test an empty candidate list.

Q Reframe it.

A Check units before combining signals, and put cuts to the list after the filters, not before.

REFERENCE CODE

```
import math

def rank_row(candidates, w1, w2, top_n, max_plays):
    def popularity(c):               # map plays to a 0 to 5 scale
        return 5.0 * math.log1p(c["plays_last_week"]) / math.log1p(max_plays)

    eligible = [c for c in candidates if not c["already_watched"]]
    eligible.sort(
        key=lambda c: (w1 * popularity(c) + w2 * c["predicted_rating"], c["title_id"]),
        reverse=True,
    )
    return [c["title_id"] for c in eligible[:top_n]]
```

Basis: The posts' statement about popularity is from Part 2 (source-confirmed); the faults are standard and the fix was checked by hand and not run.

PART 3

The ranking function: weights, learning to rank and reading evidence

How much the ratio of two weights decides, how a ranking model is learned, what a chart without a metric can and cannot say, and a loop the posts do not discuss.

READ THESE FIRST: CONCEPTS COVERED IN THIS PART

Ranking function	A function that takes a member and a title and returns a score, built from several signals. Items are sorted by score.
Weight ratio	The relative size of two weights. In a weighted sum, only the ratio between the weights, not their size, decides the order.
Learning to rank	Training a model, from positive and negative examples taken from history, to produce an ordering. The posts note it is central to search and ad targeting.
Pointwise and pairwise	Pointwise methods score each item alone. Pairwise methods learn from pairs and care about which of the two comes first.
Feature and model fit	A new feature shows no value if the model cannot learn it, and a stronger model shows none if the features do not exploit it.
Baseline	The simpler thing a result is compared with. A percentage improvement means little without the baseline and the metric.
Feedback loop	What the system shows decides what members play, which changes the data the next version learns from.
Popularity bias	A tendency for already popular items to be shown more, and so to become more popular.
Where to read	Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 1 (April 2012) | Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 2 (June 2012)

09 How much do the weights decide?

RANKING FUNCTION

SCENARIO

The posts' example ranking function is a weighted sum of popularity and predicted rating. A product manager asks you to pick the weights. Show with numbers how much they matter, and how Netflix says to set them.

HOW TO THINK ABOUT IT

Use two titles whose order flips, and find the weight ratio at which it does.

Q What is the function?

A f_rank(u, v) = w1 × p(v) + w2 × r(u, v) + b, with p the popularity and r the predicted rating. The bias b is constant and does not affect the order (source-confirmed).

- Titles are passed through the function and sorted in descending order of score.

Q Work an example.

A Both signals on a 0 to 5 scale. Title A: popularity 0.3, predicted rating 4.5. Title B: popularity 4.6, predicted rating 3.6. With w1 = 0.6 and w2 = 0.4: A scores 1.98 and B scores 4.20, so B ranks first.

- The numbers are invented. A has the higher predicted rating.

Q When does A win?

A When w2 × (4.5 − 3.6) is above w1 × (4.6 − 0.3), that is w2 × 0.9 above w1 × 4.3, a weight ratio w2 to w1 above 4.8.

- Only the ratio matters. Scaling both weights by 10 changes nothing.

Q How does Netflix say to set the weights?

A Two ways: sample the space of weights and let members decide through many A/B tests, which the post calls time consuming and not very cost effective, or formulate it as a learning problem: pick positive and negative examples from history and learn the weights that optimize the goal.

- The second is learning to rank. The post notes that for recommendations there is no global notion of relevance and the model must be personalized.

Q Reframe it.

A In a weighted score the ratio of the weights is the design. Learn it from data, and check that the scales of the inputs are comparable.

Basis: The function, the two ways to set weights and the personalization point are from Part 2 (source-confirmed); the two titles and weights are invented; arithmetic recomputed.

10 Which model learns the ranking?

MODELS

SCENARIO

A team has a linear ranking function and wants a stronger model. The post lists several families and warns about matching features and model. Advise them.

HOW TO THINK ABOUT IT

List the families, then say what the post warns about, and what the linear function cannot express.

Q What does the post name?

A Classifiers that can be used for ranking: logistic regression, support vector machines, neural networks and gradient boosted decision trees. Methods designed for learning to rank, such as RankSVM and RankBoost (source-confirmed).

- It also lists many other methods a practitioner should know, from elastic nets to matrix factorization.

Q What is the warning?

A There is no easy answer. The simpler the feature space, the simpler the model can be. It is easy to get trapped where a new feature does not show value because the model cannot learn it, or to conclude that a powerful model is useless because the features do not exploit it.

- So change them together, or you may reject something good.

Q What can a linear function not do?

A Express an interaction, such as popularity mattering more for a new member with few ratings and less for a member with years of ratings. It would need a feature built for that.

- A tree model or a neural network can learn it if the data supports it.

Q What would you do?

A Add features first, with a model that can use them, and compare against the linear function on the same data. Keep the linear function as a baseline you can explain.

- Check the result by an online test, not only by offline metrics.

Q Reframe it.

A Upgrade features and model together, and keep the simple version as the baseline. A negative result can mean the pair was mismatched.

Basis: The model families and the warning are from Part 2 (source-confirmed); the advice is reasoned.

11 Read the bar chart

EVIDENCE

SCENARIO

Part 2 has a bar chart titled Ranking improvement over baseline. Popularity is the baseline at zero, adding ratings reaches about a third, and adding more features and optimized models reaches about 225%. A slide says Netflix improved ranking by 225%. Say what the chart supports.

HOW TO THINK ABOUT IT

List what the chart labels, and what it leaves unsaid.

Q What does it show?

A Three configurations against a popularity-only baseline. The bar for adding ratings is about one third (roughly 32% by reading the bar), and the bar for more features and optimized models is about 225% (source-confirmed from the figure).

- The values are read from the bars, since the chart has no numbers.

Q What does it leave unsaid?

A The metric, whether the measurement is offline or in an A/B test, the data it was measured on, and which features gave the larger gain. The post says some features showed no positive effect while others improved ranking accuracy tremendously.

- A percentage of an unnamed metric is not a claim about members.

Q What does it support?

A The direction and rough size of the difference: adding ratings helps a little over popularity, and more features with a tuned model help far more.

- It supports the point that features and model quality matter more than adding predicted rating alone.

Q How would you word the slide?

A On an unspecified ranking metric, adding predicted rating improved over popularity by about a third, and adding more features and tuning the model improved by about 225%, per a chart in a 2012 post that does not name the metric.

- Name the source, the baseline and the unknowns.

Q Reframe it.

A A chart without a metric is a picture of a ratio. Quote what it labels, and say what it does not.

Basis: The chart's contents are from Part 2 (source-confirmed, read from the figure); the slide wording is reasoned.

12 Popular titles get more popular

RISK

SCENARIO

The ranking function includes popularity as an input, and popularity is computed from plays, which the ranking itself influences. A reviewer asks whether that is a problem. The posts do not discuss it.

HOW TO THINK ABOUT IT

Follow one title through two rounds, then ask what the posts' evaluation would and would not catch.

Q What is the loop?

A A popular title scores higher, is shown nearer the top, is played more, and becomes more popular. A title that fits a member better and starts less popular may never be shown high enough to prove it.

- This is this document's inference, since the posts do not discuss it.

Q Put numbers on two rounds.

A Title X has 1,000 plays and Title Y has 800, and the top slot gets 3 times the plays of the fifth. Y is 20% less popular and ranked below. After a week, X gets its extra exposure and the gap may widen well beyond 20%.

- The numbers are invented.

Q What do the posts' tools offer?

A Popularity over several windows and groups, which tracks trends and local tastes. Learned weights, which can lower the weight of popularity when it hurts. A/B tests with engagement and retention as the final judge.

- None is said to address this directly.

Q What would you add?

A A diversity measure, which the posts say they track, a share of traffic with random order to measure true preference, and a check of how concentrated plays are over time.

- The random-order idea comes from the other ranking systems in this series. The posts do not describe it.

Q Reframe it.

A When a system's output feeds its own input, look for a loop, and keep a way to see what would have happened without it.

Basis: The inputs of the ranking function and the diversity metrics are from the posts (source-confirmed); the loop and the remedies are reasoned and untested; the plays are invented.

PART 4

Evaluation and operations: offline metrics, A/B tests and serving

Computing ranking metrics by hand, what a test with thousands of members can see, what to cache, and what grows at ten times the scale.

READ THESE FIRST: CONCEPTS COVERED IN THIS PART

Mean reciprocal rank (MRR)	The average, across requests, of 1 divided by the rank of the correct item. Rank 1 gives 1.0 and rank 10 gives 0.1.
Fraction of concordant pairs (FCP)	Of all pairs of items where one is truly preferred, the share that the model orders correctly.
NDCG	Normalized discounted cumulative gain: rewards relevant items near the top and divides by the best possible ordering.
A/B test cell	One group of members receiving one variant. A test may have 2 to 20 cells compared against a control.
Standard error	For a rate p measured on n members, the square root of p(1 minus p) divided by n. A 95% margin is about 1.96 times that.
Multiple comparisons	Comparing many cells against one control at 5% each makes it likely that at least one looks significant by chance.
Cache staleness	How old a cached value can be. A popularity value cached for an hour is up to an hour old.
Offline-online gap	Offline metrics only indicate online results. The posts say the mapping is not perfect.
Where to read	Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 1 (April 2012) | Amatriain and Basilico: Netflix Recommendations, Beyond the 5 Stars, Part 2 (June 2012)

13 Compute the ranking metrics

METRICS

SCENARIO

The posts list NDCG, mean reciprocal rank and fraction of concordant pairs among the offline metrics. An interviewer asks you to compute two of them on small examples, and to say how far to trust them.

HOW TO THINK ABOUT IT

Compute MRR on three requests, then FCP on three titles, then read what the posts say about trust.

Q MRR for three requests.

A The correct item is at ranks 1, 3 and 10. Reciprocal ranks: 1.0, 0.33 and 0.1. MRR = (1 + 0.33 + 0.1) ÷ 3 = 0.48.

- It rewards the right item near the top and punishes it heavily further down.

Q FCP for three titles.

A Truth: A above B above C, which gives the pairs (A, B), (A, C) and (B, C). A model ordering B, A, C gets (A, B) wrong and the other two right: FCP = 2 ÷ 3. A model ordering A, B, C gets 3 ÷ 3.

- It measures order directly, which RMSE does not (Question 2).

Q What about the rest of the list?

A The posts list classification metrics, RMSE and other metrics for aspects such as diversity. No single metric catches every kind of regression.

- A ranking-order problem will not show in RMSE, and a diversity problem will not show in any order metric.

Q How far to trust them?

A The posts say Netflix tracks how well the offline metrics correlate with online gains in A/B tests, and that since the mapping is not perfect, offline performance is used only as an indication to make informed decisions on follow-up tests (source-confirmed).

- An offline gain is a reason to test, not to ship.

Q Reframe it.

A Learn the metrics by computing them on three items. Then decide what each is blind to, and keep a basket.

Basis: The list of metrics and the statement about the offline-online mapping are from Part 2 (source-confirmed); the examples are invented and the metric definitions are standard; arithmetic recomputed.

14 What can a test with thousands of members see?

EXPERIMENTS

SCENARIO

The posts say tests usually have thousands of members and 2 to 20 cells, and that scores of tests run in parallel, with engagement and retention as the trusted metrics. A colleague launches a 20-cell test and ships whichever cell has a significant gain. Examine it.

HOW TO THINK ABOUT IT

Compute the margin on a retention difference for 5,000 members a cell, then the chance of a false winner among 19 comparisons.

Q How precise is a retention number?

A With retention near 90% and 5,000 members a cell, the standard error is sqrt(0.9 × 0.1 ÷ 5,000) = 0.42%. The 95% margin is about 0.8 points.

- The cell size is an assumption inside the posts' thousands.

Q What about a difference between two cells?

A 1.96 × 0.42% × sqrt(2) = about 1.2 points. A retention change under about a point cannot be seen reliably.

- That is a large effect for a ranking change, so a more sensitive measure, such as hours of play, helps.

Q What does a 20-cell test do?

A Compared against one control at 5% each, 19 comparisons give a 1 − 0.95^19 = 62% chance that at least one shows a significant gain with no real effect.

- With 5 cells, which is 4 comparisons, it is 1 − 0.95^4 = 19%.

Q What do the posts say?

A They say to think about dependent and independent variables, control and significance, and do not describe a correction for many comparisons.

- Common remedies are a stricter threshold, fewer cells, or confirming a winner in a second test.

Q Reframe it.

A More cells mean more chances for luck. Decide the threshold and the confirmation step before the test starts.

Basis: The test sizes and the metrics are from Part 2 (source-confirmed); the 5,000 members and 90% retention are illustrative; arithmetic recomputed; the remedies are standard.

15 What can be cached?

SERVING

SCENARIO

A request must produce an ordered row for one member in real time. The posts say ranking is scoring, sorting and filtering, and give no architecture. Decide what to cache and what to compute.

HOW TO THINK ABOUT IT

Separate what is shared from what is personal, and decide how stale each can be.

Q What is personal?

A The predicted rating r(u, v) for a member and a title, and the member's history. The order is therefore different for each member and cannot be a global list.

- The posts say a global notion of relevance is not expected.

Q What is shared?

A Popularity, computed over windows and groups, and metadata. A popularity value can be cached for its window and read cheaply.

- An hourly value cached for an hour is up to an hour stale.

Q What is the cost of caching it?

A A title that spikes inside the window shows the spike late. With a spike of 63 times the weekly average (Question 6), an hour of staleness delays the signal for a title that matters.

- Shorter windows for the signals that move quickly cost more computation.

Q What is the serving path?

A Assemble candidates for the row, score each one, sort, filter the ineligible titles, and return the top. The filter step and its reasons are this document's design.

- At the assumed peak of 4.8 million scores a second, the arithmetic is small and the reads are the cost.

Q Reframe it.

A Cache the shared and slow-changing parts, compute the personal parts at request time, and state how stale each cached part can be.

Basis: The scoring, sorting and filtering framing is from Part 1 (source-confirmed); the cache design is this document's and the posts give no architecture; arithmetic recomputed.

16 Risks and ten times the scale

RISK

SCENARIO

A launch review asks what can go wrong with this ranking and what grows at ten times the members and plays. Use the posts and the sizing.

HOW TO THINK ABOUT IT

List the failure modes the posts name or imply, then scale each quantity.

Q What can go wrong?

A An offline gain that does not move engagement, a model that lets predicted rating dominate again and brings back niche picks, and a feedback loop through popularity (reasoned).

- Position effects in the presentations data are another risk the posts do not discuss.

Q What does ten times do to scoring?

A From 4.8 million to 48 million scores a second at the assumed peak. The arithmetic stays small and the number of reads grows ten times.

- More rows or titles per row would multiply it further.

Q What does it do to popularity?

A One computation per title per window per group. More members allows more groups, which multiplies the work.

- It is bounded by the number of titles, not members.

Q What does it do to tests?

A Ten times the members per cell cuts the standard error by 3.2 times, so smaller effects are visible. More parallel tests need more care about false winners.

- Question 14 shows the arithmetic.

Q Reframe it.

A Sort the quantities by what they scale with: members, titles or tests. Each needs a different response.

Basis: The failure modes the posts name are from Part 2 (source-confirmed); the others are reasoned; the 10 times figures are illustrative arithmetic.

Sources and confidence

The case study this book is built on is the AI System Design Primer's Capstone #8. The sources are the two Netflix Tech Blog posts of 2012, read in full on 2026-10-17 from archived copies, because the original addresses no longer resolve. They are company blog posts. They give the Prize figures, a handful of 2012 volumes, one function form and a chart without a stated metric, and they describe no serving architecture, so answers about serving are design arguments made for this book and were not run on real systems.

- Amatriain and Basilico: Netflix Recommendations: Beyond the 5 Stars, Part 1, Netflix Tech Blog, April 6, 2012 (archived copy, read in full 2026-10-17)
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https://web.archive.org/web/20170430194230/http://techblog.netflix.com:80/2012/06/netflix-recommendations-beyond-5-stars.html